

Writing out the first law of thermodynamics:

$$\underbrace{Q_{in} - Q_{out}}_{\left(\begin{array}{c} \text{Energy conducted} \\ \text{into the CV} \end{array} \right)} + \underbrace{Q_{gen}}_{\left(\begin{array}{c} \text{Energy} \\ \text{Generated} \\ \text{Within the} \\ \text{CV} \end{array} \right)} = \underbrace{Q_{ST}}_{\left(\begin{array}{c} \text{The increase in energy in the CV} \\ \text{per unit time} \end{array} \right)}$$

Entering & leaving the surfaces we have:

$$Q_{in} = q_x A_x + q_y A_y + q_z A_z$$

$$Q_{out} = q_{x+\Delta x} A_{x+\Delta x} + q_{y+\Delta y} A_{y+\Delta y} + q_{z+\Delta z} A_{z+\Delta z}$$

For our cartesian coordinate system:

$$A_x = A_{x+\Delta x} = \Delta y \Delta z$$

$$A_y = A_{y+\Delta y} = \Delta x \Delta z$$

$$A_z = A_{z+\Delta z} = \Delta x \Delta y$$

Within our CV, we have energy generation, Q'''

$$Q_{gen} = Q''' \forall = Q''' \Delta x \Delta y \Delta z$$

Within the control volume, the change in stored energy is:

$$Q_{ST} = \frac{dU}{dt} = \frac{d(Mu)}{dt} = \frac{d(\rho \forall u)}{dt}$$

U = internal energy
 u = specific internal energy

M = total mass of CV
 ρ = density

Assuming $p \equiv \text{constant}$

$$Q_{ST} = \rho \frac{du}{dt} \Delta x \Delta y \Delta z$$

But we know from thermodynamics that:

$$U = M c_p (T - T_{ref}) \Rightarrow \text{Equation of state for a simple incompressible substance with constant specific heat.}$$

or

$$u = c_p (T - T_{ref})$$

Back substituting:

$$Q_{ST} = \rho \frac{d(c_p (T - T_{ref}))}{dt} \Delta x \Delta y \Delta z$$

$T_{ref} = \text{constant}$ & $c_p = \text{constant}$, hence

$$Q_{ST} = \rho c_p \frac{dT}{dt} \Delta x \Delta y \Delta z$$

Putting everything together, we obtain:

$$Q_{in} - Q_{out} = \Delta y \Delta z (q_x - q_{x+\Delta x}) + \Delta x \Delta z (q_y - q_{y+\Delta y}) + \Delta x \Delta y (q_z - q_{z+\Delta z})$$

Using Taylor series expansion:

$$q_{x+\Delta x} = q_x + \frac{dq_x}{dx} \Delta x + \frac{1}{2} \frac{d^2 q_x}{dx^2} \Delta x^2 + \dots + \frac{1}{n!} \frac{d^n q_x}{dx^n} \Delta x^n$$

So

$$(q_x - q_{x+\Delta x}) = -\frac{dq_x}{dx} \Delta x - \frac{1}{2} \frac{d^2 q_x}{dx^2} \Delta x^2 - \dots - \frac{1}{n!} \frac{d^n q_x}{dx^n} \Delta x^n$$

$$(q_y - q_{y+\Delta y}) = -\frac{dq_y}{dy} \Delta y - \frac{1}{2} \frac{d^2 q_y}{dy^2} \Delta y^2 - \dots - \frac{1}{n!} \frac{d^n q_y}{dy^n} \Delta y^n$$

Same for $(q_z - q_{z+\Delta z})$.

Now let's put everything together

$$\begin{aligned}
 Q_{in} - Q_{out} = & -\Delta y \Delta z \left(\frac{dq_x}{dx} \Delta x + \frac{d^2 q_x}{2 dx^2} \Delta x^2 + \dots + \frac{1}{n!} \frac{d^n q_x}{dx^n} \Delta x^n \right) \\
 & - \Delta x \Delta z \left(\frac{dq_y}{dy} \Delta y + \frac{d^2 q_y}{2 dy^2} \Delta y^2 + \dots + \frac{1}{n!} \frac{d^n q_y}{dy^n} \Delta y^n \right) \\
 & - \Delta x \Delta y \left(\frac{dq_z}{dz} \Delta z + \frac{d^2 q_z}{2 dz^2} \Delta z^2 + \dots + \frac{1}{n!} \frac{d^n q_z}{dz^n} \Delta z^n \right)
 \end{aligned}$$

↳ Don't be scared, this will simplify a lot!

$$\begin{aligned}
 Q_{in} - Q_{out} + Q_{gen} = Q_{st} \\
 -\Delta x \Delta y \Delta z \left[\left(\frac{dq_x}{dx} + \frac{1}{2} \frac{d^2 q_x}{dx^2} \Delta x + \dots \right) + \left(\frac{dq_y}{dy} + \frac{1}{2} \frac{d^2 q_y}{dy^2} \Delta y + \dots \right) \right. \\
 \left. + \left(\frac{dq_z}{dz} + \frac{1}{2} \frac{d^2 q_z}{dz^2} \Delta z + \dots \right) \right] + Q''' \Delta x \Delta y \Delta z = \rho C_p \frac{dT}{dt} \Delta x \Delta y \Delta z
 \end{aligned}$$

Note, we can divide both sides by $\Delta x \Delta y \Delta z$ to simplify

$$\begin{aligned}
 \left(\frac{dq_x}{dx} + \frac{1}{2} \frac{d^2 q_x}{dx^2} \Delta x + \dots \right) + \left(\frac{dq_y}{dy} + \frac{1}{2} \frac{d^2 q_y}{dy^2} \Delta y + \dots \right) + \left(\frac{dq_z}{dz} + \frac{1}{2} \frac{d^2 q_z}{dz^2} \Delta z + \dots \right) \\
 + Q''' = \rho C_p \frac{dT}{dt}
 \end{aligned}$$

Taking the limit of $\Delta x, \Delta y, \Delta z \rightarrow 0$ (differential volume) many of our terms drop out, & we are left with only:

$$- \left(\frac{dq_x}{dx} + \frac{dq_y}{dy} + \frac{dq_z}{dz} \right) + Q''' = \rho C_p \frac{dT}{dt}$$

But we have already shown that:

$$q_x = -k \frac{dT}{dx} ; \quad q_y = -k \frac{dT}{dy} ; \quad q_z = -k \frac{dT}{dz}$$

Back substituting, we obtain:

$$\frac{d}{dx} \left(k \frac{dT}{dx} \right) + \frac{d}{dy} \left(k \frac{dT}{dy} \right) + \frac{d}{dz} \left(k \frac{dT}{dz} \right) + Q''' = \rho c_p \frac{dT}{dt}$$

Note though, more rigorously written, we should have $d = \partial$ since $T = f(x, y, z, t)$

$$\frac{dT}{dx} \Rightarrow \frac{\partial T}{\partial x} \text{ (partial derivative)}$$

↳ Would be valid if $T = f(x)$ only.

Rewriting, we obtain:

$$\boxed{\frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(k \frac{\partial T}{\partial y} \right) + \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right) + Q''' = \rho c_p \frac{\partial T}{\partial t}}$$

↳ Second order partial differential equation (PDE). Represents the conservation of thermal energy for an isotropic, incompressible substance, with density & specific heat independent of time.

A well posed problem requires two boundary conditions in each coordinate & an initial condition.

If thermal conductivity is isotropic (doesn't depend on location or direction):

$$\boxed{\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} + \frac{Q'''}{k} = \frac{1}{\alpha} \frac{\partial T}{\partial t}}$$

where $\alpha = \frac{k}{\rho c_p} \Rightarrow$ Thermal diffusivity material property.

For constant properties with 1D conduction:

$$\boxed{\frac{\partial^2 T}{\partial x^2} + \frac{Q'''}{k} = \frac{1}{\alpha} \frac{\partial T}{\partial t}}$$

For no heat generation, 1D conduction:

$$\boxed{\frac{\partial^2 T}{\partial x^2} = \frac{1}{\alpha} \frac{\partial T}{\partial t}}$$

For steady state, 1D, & no heat generation:

$$\boxed{\frac{\partial^2 T}{\partial x^2} = 0} \Rightarrow \text{This is probably the most useful form to solve problems.}$$

So far, we've only used cartesian coordinates, but many heat transfer problems are easier to solve with cylindrical or spherical coordinate systems:

